CORE

Routing / Movement

- Know where the hook is when the system starts, then move to a target inside the workspace.
    - Require the workspace dimensions to be configured before operation.
    - Establish the hook's starting position through homing or position sensors.
    - Accept a target position using workspace coordinates.
    - Move the hook along the X, Y and Z axes as required.
    - Prevent the track configuration from changing while the gantry is moving.
    - Stop within a centimeter of the requested position.
    - Prevent commands that would send the hook outside the configured workspace.
    - Use controlled acceleration and deceleration to reduce load sway.
    - **Edge cases:**
        - Gantry loses track of its current position.
            - Solution: Encoders track data so a gap is visible
        - The user tries to move the gantry outside of its allowed workspace.
            - Solution: Encoders track its position and slow before edge is reached
        - Two movement inputs are given at the same time/conflicting commands are received.
            - Solution: Queued execution one then the next (race condition of which goes first)
        - Movement commanded before homing.
            - Solution: Finishes homing before movement will start
        - Actual position does not match its expected position (Motor desync)
            - Solution: Error is shown and homing sequence prompted
        - Pathfinding can't connect start and end points
            - Solution: Error is shown and user is prompted to either adjust environment or choose another point
        - Remote pathfinding connection is broken during move
            - Solution: Data is already sent so it will finish the move, error will show when connection is disrupted

Safety

- Stay inside the workspace, don't lift more than the limit, and include the emergency stop here. After a stop, the system should find its position again before moving.
    - Use sensors to detect the ends of each axis.
    - Emergency stop button

- Measure the lifted load and prevent movement when the weight limit is exceeded.
- The system has to reconfirm position on loss of power
- Prevent movement when a sensor, motor, or attachment reports a fault.
- Provide clear status indicators for ready, moving, stopped, and fault conditions.
- Verify safety conditions before accepting a movement command.
- Stop safely if communication with a sensor or motor is lost.
- **Edge cases:**
  - Weight sensor reports less weight than is actually attached
    - Solution: The user replaces the censor, the system can't really detect this itself so it's up to the user to recognize the issue.
  - The component catches fire.
    - Solution: Shutdown & cut power to the system, users will have to fix the issue after shutdown.
  - Lack of power delivery or too little power (including while a load is suspended)
    - Solution: Send issue/error message to user and maintain state till problem has been resolved.
  - Motor stalls while the gantry is moving.
    - Solution:
  - Hook or payload becomes disconnected while moving.
    - Solution: Stop the system and wait for recovery or new direction from the user.
  - Load exceeds the limit after it's already lifted.
    - Solution: Stop the system and stop the current work order.
  - Limit switch triggered unexpectedly.
    - Solution: Stop the system and stop the current work order.
  - The weight sensor fails completely or stops sending data.
    - Solution: Stop the system and stop the current work order.
  - The user tries to lift or lower while the gantry is moving.
    - Solution: If area mapping can accommodate space needed and payload isn't in danger of colliding with anything, then permit the action. If not, ignore the action and continue.

Attachments

- One attachment for now (the hook) that can lift, lower, and check the weight.
  - Create a standardized mounting point for the hook.
  - Allow the hook to lift and lower a payload using the Z-axis mechanism.
  - Include a load sensor to measure the attached weight.
  - Confirm that the hook is properly connected before allowing movement.
  - Design the connection so other attachments can be added in later versions.
  - **Edge cases:**
    - Attachment is not fully secured before the gantry starts moving.

        - Solution: Give User prompt: Check attachment Connection
    - Attachment becomes loose or disconnects during movement.
        - Solution: Emergency Stop activation, returns gantry to start position
    - Wrong attachment is installed for the selected task.
        - Solution: User error prompt: Use proper attachment for task
    - The system fails to detect the installed attachment.
        - Solution: User prompt is asked to check system diagnostics and connections of attachment.
    - Attachment is too heavy for the gantry.
        - Solution: User Prompt: Machine runs diagnostic, prompts user to remove weight from object being lifted.
    - Attachment collides with the frame.
        - Solution: Emergency stop, User is prompted to check frame.
    - Attachment stops working during use.
        - Solution: Diagnostic is run to check connections for attachment. User is prompted to emergency stop the gantry.
    - The user tries to operate an attachment that is not installed.
        - Solution: User is prompted to check attachment connections.

LATER

- Support multiple interchangeable attachments.
- Add scheduled and fully automated tasks.
- Develop a more advanced app or website.
- Add obstacle detection and collision avoidance.
- Add automatic sway detection and correction.
- Store reusable positions, routes, and task sequences.
- Add remote monitoring and notifications.
- Allow the system to recognize track changes automatically.
- Payload swings too far during or after movement